

PROBLEM SPACE

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

Setting B — HRI: Mercer Observation Unit Robotic Kiosk

USER

Group(s)	Characteristics	Needs
Who are the users? Are there supporting users? - Observation Unit nurses - Healthcare assistants supervising patient monitoring primary users	What are the users like? - Clinically trained - Responsible for several patients at once - Need fast, reliable information - May not be beside the kiosk during every assessment - Work in a busy, noisy environment	- Reliable collection of vital signs - Immediate notification of abnormal or missing readings - Clear indication when a sensor fails - Manual fallback when the kiosk cannot complete an assessment - Easy access to the patient's results.
secondary users - Patients, Porters, Doctors - Biomedical engineering staff - Clinical IT staff - Hospital governance personnel	Patients may vary in age, mobility, language and health literacy. Some may be anxious, confused or unfamiliar with technology. The technical staff needs clear fault and maintenance information and the governance staff need auditability and evidence of safe operation.	- Simple instructions - Clear indication of what the robot is doing - Ability to request human help - Privacy during assessment - Reassurance that the robot supports staff rather than replacing them
Goal(s)		
- Collect routine observations safely - Identify abnormal readings quickly - Reduce repetitive monitoring workload - Receive alerts when an assessment is incomplete. primary users		- Improve consistency of Observation Unit monitoring - Support timely clinical escalation - Maintain nurse oversight of all robot-assisted assessments
secondary users - Complete a guided vital-sign check safely - Understand each step of the interaction - Obtain help easily when needed SHORT-TERM		- Improve access to timely monitoring - Maintain patient autonomy, privacy and confidence - Reduce delays without reducing human care LONG-TERM

ROBOT

Task(s)			
• Guide the patient through the assessment • Collect vital signs from connected sensors • Check whether readings are complete and valid • Display simple instructions • Send results to nursing staff • Alert staff if readings are abnormal or incomplete • Request manual assistance when a sensor fails		• Support repeated observation checks • Integrate with hospital monitoring systems • Maintain logs of readings, faults and staff responses • Adapt instructions for accessibility and patient needs • Support safe escalation without making final clinical decisions.	
SHORT-TERM		LONG-TERM	
Advantages			
What tasks does the AI system perform?			
• Uses calm, simple prompts • Reassures patients during the assessment • Directs patients to request help when needed	• Reduces uncertainty by explaining each step • Uses non-threatening language and movement • Avoids presenting the robot as a replacement for staff	• Adjustable language and text size • Can repeat instructions • Can adapt pace for elderly or anxious patients	• Repeats measurements consistently • Checks whether readings are complete • Records time and device status accurately
Social skills	Usefulness	Personalization	Precise tasks
• Collects blood pressure, pulse, oxygen saturation and temperature • Flags missing or unreliable readings • Transfers data to staff for review	• Ideally remains stationary beside the vitals station • Does not move close to patients unless necessary • Uses a stable base to reduce collision risk	• Very limited or none • Does not physically move patients or equipment • May adjust screen angle or sensor position only	• Sends readings to the nurse station • Stores fault and interaction logs • Supports future EHR integration • Provides manual entry fallback when connectivity fails
Data collection with sensors	Mobility	Environment manipulation	Connection to systems